Project 3

Introduction and Research Question

Experimental design is an important part of data science because it allows researchers to compare alternatives in a structured way and evaluate whether one version performs better than another. In this project, we conducted an A/B test comparing two versions of web applications developed for data cleaning, feature engineering, exploratory data analysis (EDA), and data visualization. The purpose of this experiment was to determine whether clearer and more guided instructional text in the web application could improve task completion, efficiency, and overall user experience. Understanding whether clear instructional support improves user performance can help guide the design of more effective and user-friendly data science tools. The code, analysis output, and supporting files for this project are available in the project GitHub repository: https://github.com/shawnshi1121-byte/5243_Project3

The two versions tested were:

- Version A (Control): a version with minimal instructional support https://019d9d0e-0e89-ecb3-5904-48ca9a8440b2.share.connect.posit.cloud/
- Version B (Treatment): a version with more guided instructional text https://019cf84b-3059-fe33-afe4-f28b52b2c117.share.connect.posit.cloud/

Our main research question was: *Does Version B lead to better user outcomes than Version A in terms of task completion, time spent, and perceived usability?*

We hypothesized that Version B would outperform Version A because the additional guidance and clearer instructions were expected to help users complete tasks more efficiently and with a better overall experience. More specifically, we evaluated the following hypotheses:

- H1: Users of Version B will complete tasks more quickly than users of Version A.
- H2: Users of Version B will report higher ratings for ease of use, clarity, and guidance than users of Version A.
- H3: Users of Version B will achieve higher task completion outcomes than users of Version A.
- H4: Users of Version B will have a higher overall usability index than users of Version A.

This aligns with the project goal of designing and evaluating an experiment with clear control and treatment groups, measurable outcomes, and statistical analysis.

Experimental Design and Methodology

This study used an A/B testing framework in which participants were randomly assigned to one of two versions of the same Shiny-based web application. Version A served as the control, and Version B served as the treatment. Both versions were designed to help users work with a dataset, apply cleaning steps, perform feature engineering, conduct exploratory data analysis, and export processed results. In both versions, users could choose a built-in demo dataset or upload a dataset, apply cleaning options such as

missing-value and outlier handling, create derived features, inspect summary outputs, and download the processed data.

The main difference between the two versions was the level of instructional support and interface guidance. Version A provided a more minimal interface with basic workflow cues and fewer explanatory details. Version B provided a more guided and polished experience through a recommended workflow, clearer section descriptions, broader preprocessing support, and more explicit support for interactive exploratory analysis. For example, Version B highlighted a step-by-step workflow, supported multiple upload formats, included scaling options, and emphasized interactive charts and export-ready outputs

The treatment was designed to test whether clearer instructions and richer guidance would improve user performance and usability outcomes. Participants were randomly assigned to one version of the app and were asked to use it to explore data and generate a chart or visual output. Their performance was then evaluated using measurable outcomes such as task completion, time spent, and perceived usability.

The independent variable in this experiment was the web app version. Group A used Version A, and Group B used Version B. Several dependent variables were measured, including approximate time spent on the task, recorded in seconds; ease of use, clarity, and guidance felt, each rated on a 1-to-7 scale; task completion level, coded as 0 for no progress, 1 for explored data, and 2 for generated a chart or visual; and an overall usability index, calculated as the average of the ease of use, clarity, and guidance ratings.

The design was chosen to test whether a more guided and supportive interface could improve user experience and performance. This setup was appropriate for A/B testing because both versions served the same purpose, while the treatment version changed interface-related factors that could reasonably influence user behavior. The sample sizes were approximately balanced across the two groups, supporting a fair comparison between Version A and Version B.

Data Collection

Data for this study was collected through real user testing. To carry out the A/B test, we shared the two web app versions with people around us and asked them to try the app and complete a Google Form after their interaction. Participants were randomly assigned to one of the two versions by receiving one app link and one matching Google Form. This helped create a fair comparison between the control group and the treatment group, which is an important part of a well-designed A/B test.

Two raw datasets were collected: App_Ver_A_Data_raw.csv for Version A and App_Ver_B_Data_raw.csv for Version B. Each dataset contains 35 observations, for a total of 70 participants. The two datasets have the same variable structure, which makes it possible to compare the two groups directly. The collected data includes approximate time spent on the task, ratings for ease of use, clarity, and guidance, and a task completion outcome. These measures were chosen because they matched the goals of the experiment and the key outcomes of interest in this A/B test.

Since the data came from user-entered responses, some cleaning was needed before analysis. The most important cleaning step involved the time spent variable. Some users entered time in inconsistent ways. For example, a few responses included units such as “minutes” or “seconds,” and some participants likely

entered the number of minutes instead of seconds. To make the data consistent, our team used a simple cleaning rule: if a recorded time value was smaller than 15, it was multiplied by 60. This rule was based on the assumption that very small values such as 2, 3, or 5 were more likely to represent minutes than total time in seconds for completing the app task.

This cleaning step was reasonable because it improved consistency across responses and reduced the effect of obvious input errors. After cleaning, the time variable better reflected comparable task duration across participants in both groups. Other variables were also checked to make sure the response format was consistent across the two datasets, and the same variable definitions were used for both versions of the app. This helped preserve data quality and made the final dataset more reliable for statistical analysis.

Overall, the data collection process followed the main goals of the project: collecting original experimental data, documenting the process clearly, and preparing a clean and organized dataset for analysis. The data was collected from real participants, with balanced group sizes and a clear cleaning process, so this dataset provided a strong foundation for evaluating the effect of the treatment design.

Statistical Analysis & Results

To compare Version A and Version B, we used both summary statistics and statistical tests. Summary statistics helped us describe the average performance of each group, and statistical tests helped us decide whether the differences between the two groups were likely to be meaningful. For the continuous outcomes, we used Welch's two-sample t-test. These outcomes were time spent, ease of use, clarity, guidance felt, and the usability index. Because some of these measures came from rating scales, we also checked the results with Mann–Whitney U tests as a robustness check. For the completion outcome, the GitHub code used a chi-square test of independence to compare completion levels across the two versions.

The descriptive results showed a clear pattern in favor of Version B. Both groups had 35 participants. For time spent, the mean was 304.49 seconds in Version A and 225.11 seconds in Version B, and the medians were 300.0 and 180.0 seconds, respectively. For ease of use, the mean score increased from 4.91 in Version A to 5.89 in Version B. For clarity, the mean increased from 4.89 to 6.00. For guidance felt, the mean increased from 3.74 to 5.94. The usability index, which was defined in the GitHub code as the average of ease of use, clarity, and guidance felt, increased from 4.51 in Version A to 5.94 in Version B.

For time spent, users in Version B finished faster on average than users in Version A. The Welch t-test gave a p-value of 0.0383, which is below 0.05. The Mann–Whitney U test also gave a similar result, with a p-value of 0.0396. This means the result is statistically significant and suggests that Version B helped users complete the task more quickly. The effect size for this outcome was Cohen's d = -0.51, which shows a moderate difference, with lower time spent in Version B.

For the three usability ratings, Version B performed better on all measures. For ease of use, the Welch t-test p-value was 0.000682, and the Mann–Whitney p-value was 0.000982. For clarity, the Welch t-test p-value was 0.0000937, and the Mann–Whitney p-value was 0.000108. For guidance felt, the Welch t-test p-value was 6.40×10^{-9}, and the Mann–Whitney p-value was 5.27×10^{-8}. These results show very

strong evidence that users rated Version B as easier to use, clearer, and more supportive. Among these three measures, the biggest improvement was in guidance felt, which matches the main goal of the treatment design.

The usability index also strongly favored Version B. In the GitHub analysis, this index was computed as the average of ease of use, clarity, and guidance felt. The mean usability index was 4.51 for Version A and 5.94 for Version B. The Welch t-test gave a p-value of 1.08×10^{-7}, and the Mann–Whitney test gave a p-value of 7.20×10^{-7}. This provides strong evidence that Version B had better overall usability, not just better performance on one single rating question.

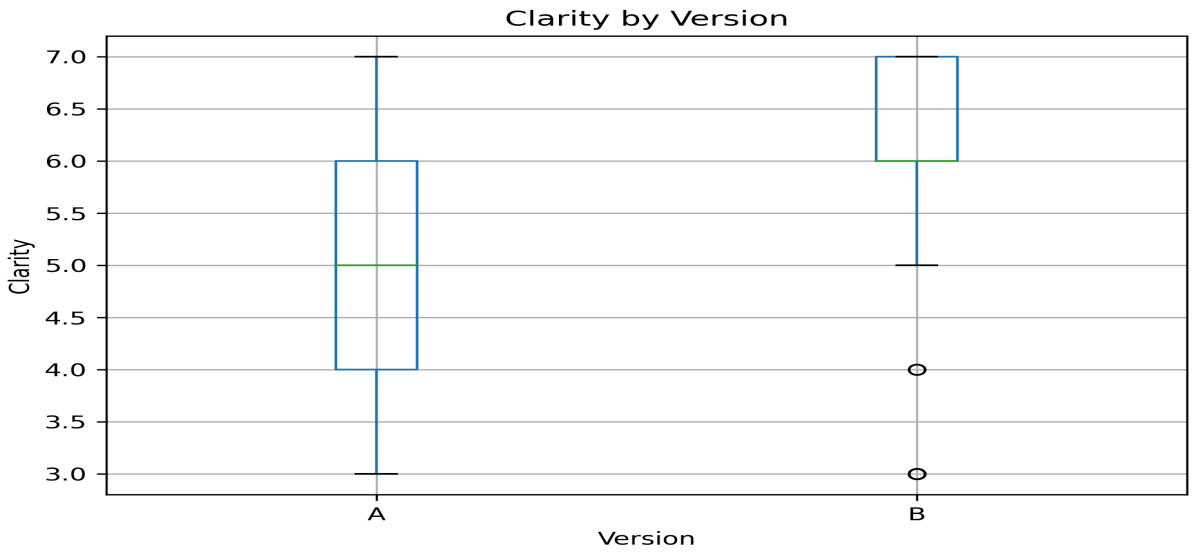
For the completion outcome, the GitHub analysis used a chi-square test on the completion levels. The test result was chi-square = 2.03 with p = 0.1547 and 1 degree of freedom. This means the completion result was not statistically significant at the 0.05 level. So, while Version B may still appear better in completion descriptively, the statistical evidence in the GitHub results is weaker for this measure than for time and the usability ratings. The reported Cramér's V = 0.170 suggests a small association between version and completion level.

Overall, the statistical results support the conclusion that Version B performed better than Version A on most important outcomes. Users in Version B finished the task faster and gave clearly higher ratings for ease of use, clarity, and guidance. The overall usability index was also much higher in Version B. The completion outcome pointed in the same direction, but the statistical evidence for completion was weaker and did not reach the usual 0.05 significance level. Taken together, these results suggest that the more guided design in Version B improved efficiency and user experience.

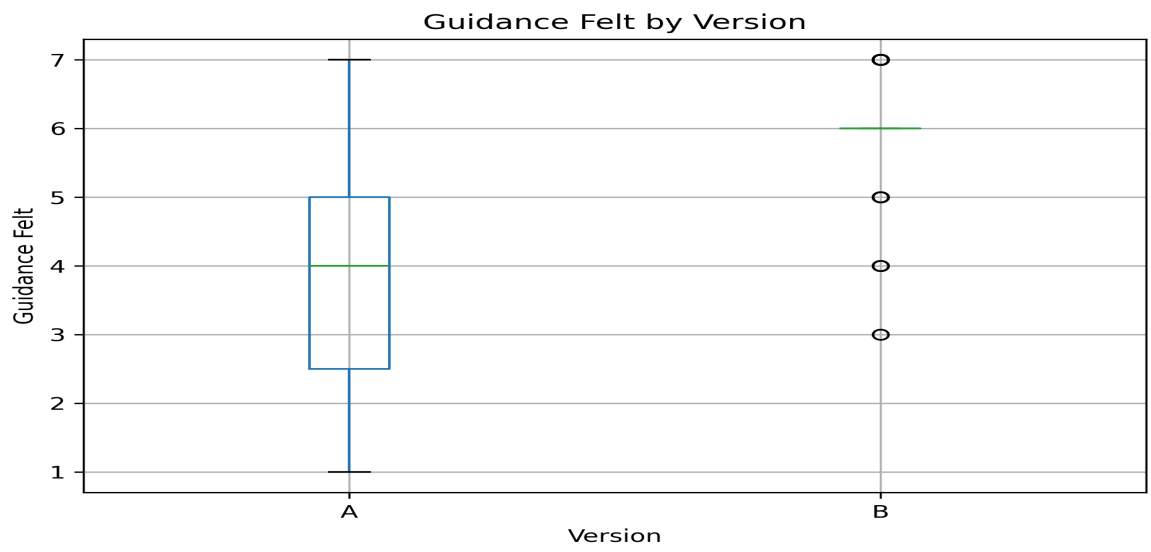
Summary Table:

Metric	**Control (A) Mean**	**Treatment (B) Mean**	**Median A**	**Median B**	**p-value (Welch)**	**Effect Size (Cohen's d)**
Completion (Chi-square)	–	–	–	–	0.155	Cramer's V = 0.17
Time (sec)	304.49	225.11	300	180	0.038	-0.51
Ease of Use	4.91	5.89	5	6	0.00068	0.85
Clarity	4.89	6.00	5	6	< 0.001	0.99
Guidance Felt	3.74	5.94	4	6	< 0.001	1.65
Usability Index	4.51	5.94	4.33	6	< 0.001	1.43

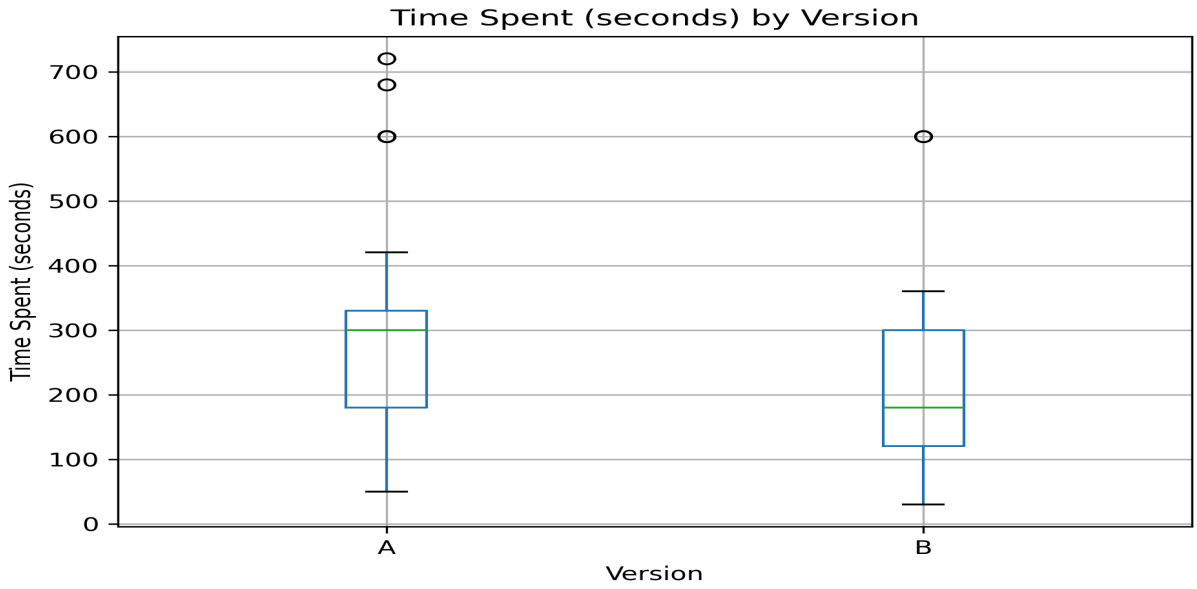
Box Plots:



The clarity-by-version box plot shows that users rated Version B as clearer than Version A, with higher scores overall and a stronger concentration of responses near the top of the scale.



The guidance-felt-by-version box plot shows that users felt much more guided when using Version B than Version A, with clearly higher ratings and a larger shift toward the upper end of the scale.



The time-spent-by-version box plot shows that users in Version B generally completed the task faster than users in Version A, with lower times overall and a lower median.

Interpretation and Conclusion

The goal of this experiment was to test whether Version B gave better user outcomes than Version A. Based on the results, the overall answer is yes. Users in Version B finished the task faster, gave higher ratings for ease of use, clarity, and guidance, and had a higher overall usability index. The completion result also favored Version B, although this difference was not statistically significant at the 5% level.

These results suggest that the design changes in Version B improved the user experience in a meaningful way. The strongest improvement was in guidance, which matches the main purpose of the treatment version. This shows that clearer instructions and a more guided workflow can help users feel more supported while using the app.

Overall, Version B should be considered the better design choice. Even though both versions had the same main function, the more guided design in Version B helped users work more efficiently and rate the app more positively. This experiment shows that interface clarity and guidance can strongly affect usability and performance.

Challenges and Limitations

Although this study produced useful findings, several limitations should be noted. First, the sample size was relatively small. The study included 70 total observations, with 35 users in Version A and 35 users in Version B. This was enough to show clear differences on several measures, but it may not have been large

enough to detect smaller effects, especially for the completion outcome. A larger sample would provide more statistical power and make the results more stable.

The time spent variable was self-reported rather than automatically recorded by the app. Because of this, some responses were entered in inconsistent ways. For example, some users included written units such as “minutes” or “seconds,” and some likely entered minutes instead of seconds. We cleaned these values using a reasonable rule, but some measurement error may still remain.

While the study used two different app versions and aimed to assign users across the two conditions, the data was collected through shared links and Google Forms. This means the experiment may not have had the same level of control as a fully automated random assignment system. As a result, there is still some possibility of selection bias, even though the A/B design was appropriate for the project goals.

Another limitation is that the participants were drawn from people around us, such as classmates, friends, or shared networks. This convenience sample made the project feasible, but it may not represent a broader population of users.

Finally, the study focused on a limited set of outcome measures: time spent, completion level, and user ratings for ease of use, clarity, and guidance. These were meaningful and relevant measures, but they do not capture every part of user behavior. Additional data such as click activity, navigation paths, error rates, or automatic session logs could provide a deeper understanding of why Version B performed better than Version A.

Despite these limitations, the study still made a strong contribution. It used original user data, compared clear control and treatment conditions, and applied statistical analysis to evaluate meaningful outcomes. In this way, the project was well aligned with the goal of designing, conducting, and analyzing an experiment with real data and clear performance measures.

Team Member Contributions

- Wentao Zhong (wz2753): Statistical Analysis & Results and Insights
- Nikhil Shanbhag (nvs2128): Data Collection and Quality& Code and Reproducibility
- Zhanhang Shi (zs2741): Experimental Design & Code and Reproducibility
- Xiying Chen (xc2781): Results and Insights & Report Structure and Writing